WENDENG WANG

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EDUCATION

University of California, San Diego (UCSD) Sep.2024 - Jun.2026

Master of Electrical and Computer Engineering

GPA:3.641/4

Central South University (CSU) Sep.2019 - Jun.2023

Bachelor of Computer Science

GPA:88.69/100

Scholarships: National Scholarship (0.2%)

RESEARCH EXPERIENCE

Development of an LLM-Agent for Cross-Platform Automated EEG Analysis Apr.2026 - present

SCCN Lab in University of California, San Diego (UCSD) *San Diego, CA*

- Agent Skill Engineering: Developed and encapsulated standardized EEG analysis "skills" within an open Agent framework, enabling the LLM to seamlessly orchestrate underlying high-performance signal processing algorithms via Function Calling.
- Robustness and Guardrail Validation: Formulated a rigorous evaluation framework and guardrails tailored for critical EEG contexts, minimizing LLM hallucinations and mis-triggers through edge-case testing to ensure zero-fault pipeline operations.

Real-Time EEG Stress Classification Mar.2025 - Oct,2025

SCCN Lab in University of California, San Diego (UCSD) *San Diego, CA*

- Cognitive Mechanism Exploration: Evaluated and quantified the dynamic correlations between mental stress states and independent EEG source components across diverse datasets, identifying highly robust neural features for predictive modeling.
- Real-Time Stress Prediction Integration: Seamlessly integrated the proprietary Orica-Clean real-time source separation pipeline with predictive models to construct an end-to-end, low-latency framework for closed-loop EEG stress forecasting.

Orica-Clean: A Python Package for Real-Time EEG Artifact Removal Jul.2025 - Jun.2026

SCCN Lab in University of California, San Diego (UCSD) *San Diego, CA*

- Implemented and Validated ORICA in Python: Replicated the Online Recursive ICA (ORICA) algorithm in Python, validating its convergence and mathematical performance against theoretical limits and standard MATLAB implementations.
- Developed Orica-Clean (Real-Time Pipeline): Engineered a production-ready Python package integrating ASR, ORICA, and an extended ICLabel to achieve real-time EEG artifact removal, source classification, and dynamic visualization.
- Validated Across Multiple Datasets: Conducted cross-scenario validation of the entire pipeline across multiple independent EEG datasets, comprehensively evaluating its real-world denoising performance, robustness, and generalization capabilities under diverse subject profiles and complex noise environments.

CNN-Based Framework for Automated Athlete Motion Classification AMA Jun.2022 - Jun.2023

Shootz, Central South University *Changsha, China*

- A tool was developed for key-frame extraction and human annotation in sports videos.**Sports Annotation**

- This study developed an automated basketball video classification framework integrating a ResNet18-based binary action classifier, a YOLOv5-based athlete detection model, and an enhanced ReID-based person re-identification module for consistent recognition across temporal segments.
- Based on the preceding processes of data collection, analysis, model training, and architectural design, this project subsequently evolved into the core technological foundation of a startup company- **Shootz**.

PUBLICATIONS

- **W. Wang**, S.-H. Hsu, T.-P. Jung, and Y. Wu, "A Closed-Loop Real-Time EEG Framework with Adaptive Denoising and Online Learning for Emotion Classification," *IEEE EMBS Neural Engineering Conference (NER)*, San Diego, CA, 2025. **(Accepted Poster Presentation)**

WORKING EXPERIENCE

Research Assistant Feb.2025 - present
SCCN Lab, University of California, San Diego *San Diego, CA*

- Implemented a Python real-time EEG artifact removal system using ORICA and ICLabel.
- Worked on real-time stress and emotion detection based on EEG signals.
- Assisted with EEG data collection in movement and VR environments.

Machine Learning Engineer (Core Developer) Jun.2022 - Sep.2023
Shootz, Central South University *ChangSha, China*

- Led the development of a data annotation tool and automation framework for large-scale visual datasets.
- Responsible for dataset labeling, deep learning model training for computer vision, and demo implementation as a core project member.

Software Development Engineer Sep.2023 - Feb.2024
ZValley Technology *ChangSha, China*

- Contributed to the development and testing of the SCRM system, a mobile trading marketing app designed to streamline user transactions.

SKILL

- **Software Skills:** Ptython / Java / C / Matlab / R
- **EEG packages:** EEGLab / BCILab / MNE / Meegkit
- **Machine Learning:** PyTorch / TensorFlow / ResNet / Yolo / EEGNet ..